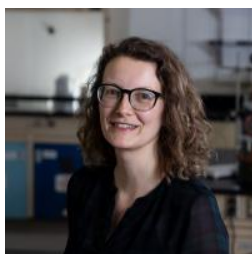


Chemistry and Chemical Biology
CHEM 1AA3
Introductory Chemistry II
Winter 2027



SCIENCE

Instructor Information



Dr. Linda Davis (she/her) - Course Coordinator

Email: genchem@mcmaster.ca

Office: ABB 121

Scheduling, permission, and absences for labs or tests should be addressed to the Course Coordinator.



Dr. Lydia Chen

Email: chenl109@mcmaster.ca

Office: See Chem Help Desk Hours posted on Avenue



Dr. Sharonna Greenberg

Email: greenbsh@mcmaster.ca

Office: See Chem Help Desk Hours posted on Avenue

Course Information

These information sheets apply to all sections of this course (C01-C04).

COURSE DESCRIPTION

A discussion of organic chemistry, chemical kinetics and acid-base equilibrium, with emphasis on relevant experimental techniques and solving real problems ranging from drug discovery to environmental chemistry.

Lecture days/times and locations are scheduled into your MOSAIC timetable. Recordings of lectures are found on the Echo360 platform available through our course Avenue page.

WHMIS 1A00

All students taking chemistry courses must complete (or have previously completed) this safety course presented by Environmental & Occupational Health Support Services (EOHSS). Please register through Mosaic. The course will then appear as a module (with pass/fail test) in Avenue.

Course Dates: 01/04/2027 - 04/07/2027

Units: 3.00

Course Delivery Mode: In Person

Course Description: A discussion of organic chemistry, chemical kinetics and acid-base equilibrium, with emphasis on relevant experimental techniques and solving real problems ranging from drug discovery to environmental chemistry.

Three lectures, one lab (two and one half hours) every other week; one term

Prerequisite(s): CHEM 1A03 or CHEM 1E03 Not open to students with credit or registration in ISCI 1A24 A/B.

This section is a guideline of topics covered in this course. Please note that a set of learning objectives for each unit will be made available on Avenue before each major assessment and further details the topics covered for each unit. Note that laboratory experiments are also a formal part of the course content and will be included on tests and exams.

CHEM 1AA3 Course Topics

Topics
Unit 1: Additional Aspects of Acid-Base Equilibrium
Unit 2: Intermolecular Forces (IMF)
Unit 3: Kinetics
Unit 4a: Organic Structure and Bonding
Unit 4b: Organic Reactivity
Unit 5: Chemical Biology

CHEM 1AA3 Laboratory Experiments

Experiment Number and Title

Experiment Number and Title
Experiment 6: Titrations
Experiment 7: Determination of the Isoelectric Point of Casein
Experiment 8: The Rate of a Chemical Reaction
Experiment 9: Organic Chemistry Part I
Experiment 10: Organic Chemistry Part II

Important Links

- [Mosaic](#)
- [Avenue to Learn](#)
- [Student Accessibility Services - Accommodations](#)
- [McMaster University Library](#)
- [eReserves](#)
- [Open Education Resources](#)

Course Learning Outcomes

- Discuss chemical concepts, theories and examples of fundamental chemistry
- Apply chemistry to current examples within the themes of health, energy, the environment and materials
- Help develop skills needed to solve chemical problems
- Provide simulated experience in experimental investigation of chemical questions

Required Materials and Texts

Please sign in with your MacID [here](#) to view your booklist

Chemistry 2e from OpenStax

ISBN: 978-1-947172-62-3

Authors: Flowers et al.

Publication Date: 2020

Edition: 2nd Ed

- [View online](#) (Links to an external site.)
- [Download a PDF](#) (Links to an external site.)

- [Order a print copy](#) (Links to an external site.)

Lab Manual +Stemle Bundle

Available through Immediate Access. Further details posted to Avenue.

Lab Safety Goggles

Available at the Campus Store

Lab Coat

Available at the Campus Store

Calculator; CASIO FX 991 MS or MS PLUS

Available at the Campus Store

Optional Course Materials

Please sign in with your MacID [here](#) to view your booklist

iClicker REEF Polling – details posted on Avenue before classes begin

Class Format

In Person

In addition to the in-person components, this course will use the following platforms to deliver course material and manage virtual meetings

- **Avenue 2 Learn** ('Avenue' or 'A2L'), an integrated set of tools for delivering course components over the Internet. This is our main platform for communicating to students and should be checked daily. Details on accessing this platform are provided below.
- **Microsoft Teams** ('MS Teams' or 'Teams'); some instructor, coordinator and TA office hours will be delivered through the Teams platform. Every student is provided with an Office 365 subscription, which includes MS Teams. Links and other details about MS Teams will be communicated to students via Avenue at the start of term.
- **Echo360** will be used to record in-person lectures. Students can access these recordings via links on our Avenue course page
- **Stemle Learning Inc** is used for practice homework assignments as well as submission of for-credit post-lab assignments for all laboratory experiments.

If students taking online assessments experience technical difficulties, the student must immediately report the difficulty. Reporting may take different forms dependent upon the online platform being used. Options to report difficulties may include emailing genchem@mcmaster.ca or their designate; accessing a chat function (where available) within the online platform to document any difficulties.

Course Evaluation

Marking Scheme

Three weighting options are shown below. Each student will automatically receive the highest grade of the three calculated grades.

Marking Scheme for CHEM 1AA3

Course Component	Option 1	Option 2	Option 3	Notes
iClicker Polling	2%	2%	2%	Optional. See Note 1.
Avenue Quizzes	10%	10%	10%	Lowest quiz mark dropped
Labs	15%	15%	15%	Mandatory. See Note 2.
Term Test 1	20%	0%	20%	See Note 3
Term Test 2	20%	20%	0%	Cumulative. See Note 3.
Final Exam	33%	53%	53%	Cumulative. Mandatory.

- **Note 1:** Students who do not participate in, or perform poorly on iClicker polls relative to their final exam score, will have this weight moved to the final exam, automatically
- **Note 2:** Students must complete and submit a post-lab assignment for a minimum of 4 laboratory experiments to pass the course. Post-lab assignments that are found to be in violation of McMaster's academic integrity policy are not counted as a submission.
 - **Students with an approved laboratory exemption** will have the laboratory component removed (0% weight). Under Option 1, the 15% laboratory weight will be redistributed equally among Term Test 1, Term Test 2, and the Final Exam, increasing each

component by 5%. Options 2 and 3 will then be adjusted accordingly to reflect this redistribution.

- **Note 3:** To obtain credit for CHEM 1AA3, students must complete a minimum of 75% of the weight of the course work as shown in Option 1. This 75% must include both the final exam and laboratory components. Students who miss both Test 1 and Test 2 will not be able to meet this 75% threshold.

Course Evaluation Details

iClicker Polling

Instructors will use the iClicker classroom response system for in-class questions. Participation in iClicker is optional, but highly recommended and can be worth up to 2% of your final grade (see Course Evaluation for more details). Each for-credit iClicker polling session/lecture is graded out of 10.00, with 5.00 marks for participation and 5.00 for the overall correctness of answers. iClicker polls taken in class before the add/drop date will not be included in the final score determination. iClicker scores will be finalized at the end of term. iClicker scores are counted as follows:

iClicker Mark (%)	iClicker Score (out of 2)
80-100	2.00
65-79	1.60
50-64	1.30
40-49	1.0
<40	0.0

Quizzes

Six quizzes are available during the term through Avenue. Each quiz will be available a minimum of two days before its due date (posted in the course schedule). All students are pre-emptively granted a one-week extension on these quizzes, except for the final quiz which will close after one week or on the last day of registrar-scheduled classes (whichever comes first). You will have two attempts at each quiz where the first attempt is the full set of questions and the second is an opportunity to rework incorrect responses for

additional credit. Only the top 5 of the 6 quiz marks will be counted towards your final grade.

Laboratories

There are 5 experiments scheduled throughout the term. Each experiment is worth 3% of your final grade and is comprised of pre- and post-lab work.

Listed below. Pre- and post-lab assessments are available on Stemble.

Lab Component	Submission Guidelines and Notes	Grade Notes
Exp # Pre-lab Quiz	Complete this well before your scheduled lab time begins. Students who have not achieved a passing grade on the pre-lab quiz will not be granted entry to the lab.	Each prelab quiz is graded within the Stemble platform and worth 1.5 % of your final grade.
Exp # Post-Lab Report	Due at 11:59 pm on the day of your scheduled lab. All students are automatically granted a 24-hour extension from this due date/time, without penalty.	Each report is graded within the Stemble platform and worth 1.5% of your final grade.

Additional notes:

- You will have multiple attempts at each pre-lab and post-lab task. The specific number of attempts for each task is posted within the platform. If no attempt counter is visible, then the task can be submitted an unlimited number of times. There is no grade deduction for subsequent attempts. The attempt with the highest score for a given task is used to determine your overall final mark on the assignment.
- Feedback is provided within the platform however it is difficult for it to predict every possible incorrect response and/or mistake. Please be advised that additional feedback is available from Stemble Support (support@stemble.ca) as well as the Avenue discussion boards.

All students must watch the Safety Video on Avenue and pass the safety quiz on Stemble. Students who complete the quiz by the deadline with a score of

100% will receive a 0.5% bonus added to the final course mark. This safety quiz is different from, and in addition to, the one completed for WHMIS 1A00.

Term Tests 1 & 2

Two 90-minute term tests are scheduled during the semester (see Course Schedule for dates and times). Test rooms will be announced on Avenue. Pre-existing conflicts should be discussed with the course coordinator (genchem@mcmaster.ca) a minimum of 1 week in advance of the test date. Please note that we will not accept any test that is started 30 minutes past the start of the official test start time on the day of the scheduled write. This includes any test written at the SAS Testing Centre so please ensure that your test start time is booked at or before the official start time on the test day shown on the Course Schedule.

Final Exam

A single 150-minute, in-person final exam is scheduled for the end of the semester by the Registrar's Office and will test all course content from the course. This examination **must be written** to pass the course. Pre-existing conflicts should be discussed with the Registrar's office once the exam timetable and locations/delivery are released. Further details about Exam scheduling and policies is found here: [Exams | Office of the Registrar | McMaster University](#)

Undergraduate Grading Scale

Grade	Equivalent Grade Point	Equivalent Percentages
A+	12	90-100
A	11	85-89
A-	10	80-84
B+	9	77-79

Grade	Equivalent Grade Point	Equivalent Percentages
B	8	73-76
B-	7	70-72
C+	6	67-69
C	5	63-66
C-	4	60-62
D+	3	57-59
D	2	53-56
D-	1	50-52
F	0	0-49

Course Schedule

COURSE SCHEDULE

Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
	January					
Week 1 No Labs	4 (Classes Begin)	5	6	7	8	9
Week 2 No Labs	11	12 Add/Drop Ends	13	14 Quiz 1 Due	15	16
Week 3 Exp 6 L1**	18	19	20	21	22	23
Week 4 Exp 6 L2**	25	26	27	28 Quiz 2 Due	29	30 Test 1, 10:00 AM
	February					
Week 5 Exp 7 L1**	1	2	3	4	5	6

Week	Monday	Tuesday	Wednesday	Thursday	Friday	Saturday
Week 6 Exp 7 L2**	8	9	10	11 Quiz 3 Due	12	13
Reading Week - February 15-21 - No Labs/Classes						
Week 7 Exp 8 L1**	22	23	24	25	26	27
	March					
Week 8 Exp 8 L2**	1	2	3	4 Quiz 4 Due	5	6
Week 9 Exp 9 L1**	8	9	10	11	12	13 Test 2, 10:00 AM
Week 10 Exp 9 L2**	15	16	17	18 Quiz 5 Due	19	20
Week 11 Exp 10 L1**	22	23	24	25	26	27
	April					
Week 12 Exp 10 L2**	29	30	31	1 Quiz 6 Due	2 Good Friday	3
Week 13 Makeup labs	5	6	7 Last day of classes	8	9	10

Laboratory Overview, Operation, and Safety

LAB SAFETY AND SCHEDULING

Dress-code for the lab is strict. All students must wear clothing to ensure their skin is covered from neck-to-wrist-to-toes. This includes:

- long pants/skirt

- a lab coat
- shoes that cover the ENTIRE foot
- safety goggles
- tie up any longhair and safely secure necklaces, scarves, or other dangling clothing or accessories

This is a fully operational chemistry lab and there is **zero-tolerance for students not conforming to safe practices**. Students who are not dressed properly for the lab will be told to leave and will earn a grade of zero for the in- and post-lab assessments.

All students are scheduled to complete 5 laboratory experiments during the term. Experiments take place every-other week with your lab day and time indicated on your Mosaic timetable as L01, L03, etc... Please note that the room listed on your Mosaic timetable may not be the room in which you complete your lab. **Final room assignments will be posted on Avenue before labs begin.**

Lab Exemptions

Any students repeating this course who would like to be exempt from the lab program must submit a request to [this form](#) posted on Avenue by **January 14th at 11:59 pm**. The criterion used for lab exemption is two-fold: completion of this course or its antirequisite at McMaster University within the last 4 years, and completion of 4 of the 5 in-person lab experiments. There is no partial exemption for some of the labs. There is no lab exemption granted on the basis of courses taken at another university. Students with an approved laboratory exemption will have the laboratory component removed (0% weight) in the Course Marking Scheme. Under Option 1, the 15% laboratory weight will be redistributed equally among Term Test 1, Term Test 2, and the Final Exam, increasing each component by 5%. Options 2 and 3 will then be adjusted accordingly to reflect this redistribution.

To ensure equity with students completing the laboratory component, students who submit an approved lab exemption request by the posted deadline will receive a 0.5-mark bonus in place of the safety quiz bonus.

Please note that lab content is still considered testable on Midterms 1 and 2 and the Final Exam. You will be responsible for ensuring you review this content.

Absences, Missed Work, Illness

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Our course is designed so that you can miss the occasional piece of work and not be penalized. MSAFs are not required to report these absences.

Component	MSAF Required?	Relief for Missed Work
iClicker Polls	No	This is an optional component. Students who score higher on their final exam relative to their iClicker score will automatically have their iClicker score dropped from their final grade calculation.

Component	MSAF Required?	Relief for Missed Work
Quizzes	No	Due dates are posted in the Course Schedule. Automatic extensions are granted for up to one week after the due date or the last day of registrar-scheduled classes (whichever comes first). There are no makeup quizzes or further extensions. Instead, we automatically drop the lowest one of the six quiz marks.
Pre-lab quizzes	No	You must submit your pre-lab quiz before the start of your lab. Failure to do so may result in your being turned away from the lab at the discretion of the TA or lab coordinator. If you are unwell or unable to attend your lab, you have until the last day of registrar classes OR a makeup lab (if eligible) to complete your pre-lab quiz.

Component	MSAF Required?	Relief for Missed Work
Post-lab report	No	Students must complete a minimum of 4 of the 5 post-lab reports to pass this course. A first missed lab report will be reweighed to the final exam. Subsequent missed labs must be completed during the makeup lab session. See MISSED/MAKEUP LAB DETAILS below, for further details on makeup/missed labs.
Term Test 1	No	There are no makeup tests. Students missing Test 1 will have its weight moved to the final exam, automatically (see Option 2 in marking scheme above). Please note that you need to write at least one of the two midterm tests to pass CHEM 1A03/1E03/1AA3

Component	MSAF Required?	Relief for Missed Work
Term Test 2	No	There are no makeup tests. Students missing Test 2 will have its weight moved to the final exam, automatically (see Option 3 in marking scheme above). Please note that you need to write at least one of the two midterm tests to pass CHEM 1A03/1E03/1AA3
Final Exam	MSAF not available	The MSAF on-line, self-reporting tool cannot be used to apply for any missed final examination or its equivalent. See <i>Petitions for Special Consideration</i> in the Undergraduate Calendar.

MISSED/MAKEUP LAB DETAILS

Pre-Lab Work

Pre-lab work must be submitted before the start of your scheduled lab period. Students submitting their work late may be turned away from the lab at the discretion of the TA or lab coordinator. If you are unable to complete your pre-lab work because you are unwell/unavailable and will not be attending your lab, then submit your pre-lab work as soon as you are well enough to do so. The final deadline to submit a pre-lab quiz is **the last day of registrar-scheduled classes at 11:59 pm.**

Work not received by this deadline will be given a grade of zero. If you plan to attend a makeup lab, but do not submit your pre-lab work by this deadline, then you will forfeit your spot in the makeup lab.

Makeup Labs

Makeup labs will be scheduled towards the end of term (details posted to Avenue). Below we outline details on eligibility to register based on the number of labs missed throughout the term. Details on how to register will be posted to Avenue.

Makeup Lab Eligibility

Number of Missed Labs	Instructions Eligibility
1	You are not eligible to register for a makeup lab. Submit your pre-lab quiz according to the instructions posted above. The weight of the flowchart and lab report will be moved to the final exam
2	Submit your pre-lab work according to the instructions posted above. Register for and attend a makeup lab for ONE of the missed labs. Submit this report as per the instructions in the lab manual. For the SECOND missed lab, the weight of the flowchart and lab report will be moved to the final exam.
3	Submit your pre-lab work according to the instructions posted above. Register for and attend/complete a makeup lab for TWO of the missed labs. Submit these reports as per the instructions in the lab manual. For the THIRD missed lab, the weight of the flowchart and lab report will be moved to the final exam.
4	You are at risk of failing CHEM 1AA3. Book an appointment with an academic advisor in your faculty office.

Number of Missed Labs	Instructions Eligibility
5	You are at risk of failing CHEM 1AA3. Book an appointment with an academic advisor in your faculty office.

Turnitin.com

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. A2L, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. All submitted work is subject to normal verification that standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Generative AI: Some Use Permitted

Students may use generative AI in this course in accordance with the guidelines outlined for each assessment, and so long as the use of generative AI is referenced and cited following citation instructions given in the syllabus. Use of generative AI outside assessment guidelines or without citation will constitute academic dishonesty. It is the student's responsibility to be clear on the limitations for use for each assessment and to be clear on the expectations for citation and reference and to do so appropriately.

<https://libguides.mcmaster.ca/cite-gen-ai>

APPROVED ADVISORY STATEMENTS

Academic Integrity

You are expected to exhibit honesty and use ethical behaviour in all aspects of the learning process. Academic credentials you earn are rooted in principles of honesty and academic integrity. **It is your responsibility to understand what constitutes academic dishonesty.**

Academic dishonesty is to knowingly act or fail to act in a way that results or could result in unearned academic credit or advantage. This behaviour can result in serious consequences, e.g. the grade of zero on an assignment, loss of credit with a notation on the transcript (notation reads: "Grade of F assigned for academic dishonesty"), and/or suspension or expulsion from the university. For information on the various types of academic dishonesty please refer to the [Academic Integrity Policy](#).

The following illustrates only three forms of academic dishonesty:

- plagiarism, e.g. the submission of work that is not one's own or for which other credit has been obtained.
- improper collaboration in group work.
- copying or using unauthorized aids in tests and examinations.

Authenticity / Plagiarism

Some courses may use a web-based service (Turnitin.com) to reveal authenticity and ownership of student submitted work. For courses using such software, students will be expected to submit their work electronically either directly to Turnitin.com or via an online learning platform (e.g. Avenue to Learn, etc.) using plagiarism detection (a service supported by Turnitin.com) so it can be checked for academic dishonesty.

Students who do not wish their work to be submitted through the plagiarism detection software must inform the Instructor before the assignment is due. No penalty will be assigned to a student who does not submit work to the plagiarism detection software. **All submitted work is subject to normal verification that**

standards of academic integrity have been upheld (e.g., on-line search, other software, etc.). For more details about McMaster's use of Turnitin.com please go to www.mcmaster.ca/academicintegrity.

Courses with an On-line Element

Some courses may use on-line elements (e.g. e-mail, Avenue to Learn, LearnLink, web pages, capa, Moodle, ThinkingCap, etc.). Students should be aware that, when they access the electronic components of a course using these elements, private information such as first and last names, user names for the McMaster e-mail accounts, and program affiliation may become apparent to all other students in the same course. The available information is dependent on the technology used. Continuation in a course that uses on-line elements will be deemed consent to this disclosure. If you have any questions or concerns about such disclosure please discuss this with the course instructor.

Online Proctoring

Some courses may use online proctoring software for tests and exams. This software may require students to turn on their video camera, present identification, monitor and record their computer activities, and/or lock/restrict their browser or other applications/software during tests or exams. This software may be required to be installed before the test/exam begins.

Conduct Expectations

As a McMaster student, you have the right to experience, and the responsibility to demonstrate, respectful and dignified interactions within all of our living, learning and working communities. These expectations are described in the [Code of Student Rights & Responsibilities](#) (the "Code"). All students share the responsibility of maintaining a positive environment for the academic and personal growth of all McMaster community members, **whether in person or online**.

It is essential that students be mindful of their interactions online, as the Code remains in effect in virtual learning environments. The Code applies to any interactions that adversely affect, disrupt, or interfere with reasonable participation in University activities. Student disruptions or behaviours that interfere with university functions on online platforms (e.g. use of Avenue 2 Learn, WebEx or Zoom for delivery), will be taken very seriously and will be investigated. Outcomes may include restriction or removal of the involved students' access to these platforms.

Academic Accommodation of Students with Disabilities

Students with disabilities who require academic accommodation must contact [Student Accessibility Services](#) (SAS) at 905-525-9140 ext. 28652 or sas@mcmaster.ca to make arrangements with a Program Coordinator. For further information, consult McMaster University's [Academic Accommodation of Students with Disabilities](#) policy.

Requests for Relief for Missed Academic Term Work

In the event of an absence for medical or other reasons, students should review and follow the [Policy on Requests for Relief for Missed Academic Term Work](#).

Academic Accommodation for Religious, Indigenous, or Spiritual Observances (RISO)

Students requiring academic accommodation based on religious, Indigenous or spiritual observances should follow the procedures set out in the [RISO](#) policy. Students should submit their request to their Faculty Office ***normally within 10 working days*** of the beginning of term in which they anticipate a need for accommodation or to the Registrar's Office prior to their examinations. Students should also contact their instructors as soon as possible to make alternative arrangements for classes, assignments, and tests.

Copyright and Recording

Students are advised that lectures, demonstrations, performances, and any other course material provided by an instructor include copyright protected works. The Copyright Act and copyright law protect every original literary, dramatic, musical and artistic work, **including lectures** by University instructors.

The recording of lectures, tutorials, or other methods of instruction may occur during a course. Recording may be done by either the instructor for the purpose of authorized distribution, or by a student for the purpose of personal study. Students should be aware that their voice and/or image may be recorded by others during the class. Please speak with the instructor if this is a concern for you.

Extreme Circumstances

The University reserves the right to change the dates and deadlines for any or all courses in extreme circumstances (e.g., severe weather, labour disruptions, etc.). Changes will be communicated through regular McMaster communication channels, such as McMaster Daily News, Avenue to Learn and/or McMaster email.